

Name: _____ Date: _____

Period: _____

Parametrically Defined Circles and Lines — Key

1. A circle is centered at $(3, -1)$ with radius 4. Write parametric equations for *counterclockwise* motion that starts at $(7, -1)$.

$$x(t) = 3 + 4 \cos t \quad y(t) = -1 + 4 \sin t$$

Domain: $0 \leq t \leq 2\pi$

2. Write a parametrization for the line segment from $(0, 6)$ to $(4, 0)$.

$$x(t) = 0 + 4t = 4t \quad y(t) = 6 + (0 - 6)t = 6 - 6t$$

Domain: $0 \leq t \leq 1$

At what value of t does the segment cross the x -axis (where $y = 0$)?

Set $y(t) = 0$: $6 - 6t = 0 \Rightarrow 6t = 6 \Rightarrow t = 1$.

The segment reaches the x -axis at $t = 1$, which is the endpoint $(4, 0)$.

Teacher Note

Q1 checks the transformation template directly. Q2 is purposely designed so the x -axis crossing is exactly the endpoint ($t = 1$). If students get $t = 1$, ask: “Is that in the domain? What point does it correspond to?” This builds the habit of checking domain validity.